1(a(i)) $v_r = 4-1$

$= 3 \text{ ms}^{-1}$

$\dot{m} = \rho V A$

$= 400 \times 3 \left(\pi \frac{(4\times10^{-2})^2}{4}\right)$

$= \frac{12}{25}\pi \text{ kgs}^{-1}$

Since the flow is frictionless and incompressible:

$V_{in} = V_{out}$

$F_x = \dot{m} V_{out} - \dot{m} V_{in}$

$= (-3\cos 60° - 3)\frac{12}{25}\pi$

$= -\frac{54}{25}\pi \text{ N} \approx -6.79\text{N}$

$F_y = \dot{m} V_{out} - \dot{m} V_{in}$

$= (3\sin 60° - 0)\frac{12}{25}\pi$

$= 3.917806627$

$\approx 3.92\text{N}$

$\therefore$ The force components are:

$F_x = 6.79\text{N} \uparrow$

$F_y = -3.92\text{N} \downarrow$

1aii) Since the flow is frictionless and incompressible:

$V_{out} = V_{in}$

$V_r = 4 + 1$

$= 5\ ms^{-1}$

$\dot{m} = \rho Q = \rho VA$

$= 400 \times 5\left(\frac{\pi(0.04)^2}{4}\right)$

$= \frac{4}{5}\pi\ kg\ s^{-1}$

$F_x = \dot{m}V_{out} - \dot{m}V_{in}$

$= (-5\cos 60° - 5)\frac{4}{5}\pi$

$= -6\pi \approx -18.85\ N$

$F_y = \dot{m}V_{out} - \dot{m}V_{in}$

$= (5\sin 60° - 0)\frac{4}{5}\pi$

$= 10.88279619\ N$

$\approx 10.88\ N$

$P = F_x v_x + F_y \cancel{v_y}^{0}$

$= 6\pi(1)$

$= 6\pi$

$\approx 18.85\ W$

1b) $\dot{m} = \rho Q = 1000 \times 1.2 \times 10^{-4}$ $\dot{m}_2 = m$

$= 0.12\,kgs^{-1}$

$$V = \frac{Q}{A} = \frac{Q}{\frac{\pi D^2}{4}} = \frac{4Q}{\pi D^2} = \frac{4(1.2\times10^{-4})}{\pi(4\times10^{-3})^2}$$

$$= 9.549296586$$

$$\approx 9.55\,ms^{-1}$$

$$T_{shaft} = \sum_{out} \dot{m}(\vec{r}\times\vec{V}) - \sum_{in} \dot{m}(\vec{r}\times\vec{V})$$

$$0 = 0.06\times0.3(9.55-0.3\omega) + 0.06\times0.2(9.55-0.2\omega)$$

$$\omega = 36.72806379$$

$$\approx 36.73\,rad\,s^{-1}$$ G

The sprinkler will rotate anti-clockwise.

2a) $V = f(\rho, D, \mu, \sigma, g)$

$V = LT^{-1}$

$\rho = ML^{-3}$

$D = L$

$\mu = ML^{-1}T^{-1}$

$\sigma = ML^{-1}T^{-2}$

$g = LT^{-2}$

Repeating variables are g, D, μ

$$\pi_1 = Vg^aD^b\mu^c = LT^{-1}L^aT^{-2a}L^bM^cL^{-c}T^{-c}$$
$$= L^{a+b-c}T^{-2a-c}M^cLT^{-1}$$
$$= L^{a+b-c+1}T^{-2a-c-1}M^c$$

$$\left.\begin{aligned} a+b-c &= -1 \\ -2a-c &= 1 \\ c &= 0 \end{aligned}\right\} \text{Solving:} \quad \begin{aligned} a &= -0.5 \\ b &= -0.5 \\ c &= 0 \end{aligned}$$

$$\therefore \pi_1 = \frac{V}{\sqrt{gD}}$$

$$\pi_2 = \rho g^aD^b\mu^c = ML^{-3}L^{a+b-c}T^{-2a-c}M^c$$
$$= L^{a+b-c-3}T^{-2a-c}M^{c-1}$$

$$\left.\begin{aligned} a+b-c &= 3 \\ -2a-c &= 0 \\ c &= 1 \end{aligned}\right\} \text{Solving:} \quad \begin{aligned} a &= -0.5 \\ b &= 4.5 \\ c &= 1 \end{aligned}$$

$$\therefore \pi_2 = \frac{\rho D^{4.5}\mu}{\sqrt{g}}$$

2a) $\pi_3 = \sigma g^a D^b \mu^c = ML^{-1}T^{-2} L^{a+b-c} T^{-2a-c} M^c$
$= L^{a+b-c-1} T^{-2a-c-2} M^{c+1}$

$$\left.\begin{aligned} a+b-c &= 1 \\ -2a-c &= 2 \\ c &= -1 \end{aligned}\right\} \quad \text{Solving:} \quad \begin{aligned} a &= -0.5 \\ b &= 0.5 \\ c &= -1 \end{aligned}$$

$$\therefore \pi_3 = \frac{\sigma\sqrt{D}}{\mu\sqrt{g}}$$

$$\therefore \frac{V}{\sqrt{gD}} = f\left(\frac{\rho D^{4.5}\mu}{\sqrt{g}}, \frac{\sigma\sqrt{D}}{\mu\sqrt{g}}\right)$$

b)

	D	V	μ	ρ	σ
SAE oil	9×10^{-5}	8.103×10^{-5}	0.287	878	0.025
Olive oil			0.084	917	0.032

$$\left(\frac{\rho D^{4.5}\mu}{\cancel{\sqrt{g}}}\right)_{SAE} = \left(\frac{\rho D^{4.5}\mu}{\cancel{\sqrt{g}}}\right)_{Olive}$$

$$878(9\times10^{-5})^{4.5}(0.287) = 917D^{4.5}(0.084)$$
$$D^{4.5} = 2.04\times10^{-18}$$
$$D = 1.17\times10^{-4}\,m$$

$$\left(\frac{V}{\sqrt{gD}}\right)_{SAE} = \left(\frac{V}{\sqrt{gD}}\right)_{Olive}$$

$$\frac{8.103\times10^{-5}}{\sqrt{9.81(9\times10^{-5})}} = \frac{V}{\sqrt{9.81(1.17\times10^{-4})}} \rightarrow V = 9.24\times10^{-5}\,ms^{-1}$$

3a) $\varepsilon = 1.5\,mm$ $\quad \nu = 1\,mm^2s^{-1} = 10^{-6}\,m^2s^{-1}$

$\mu = \rho\nu$

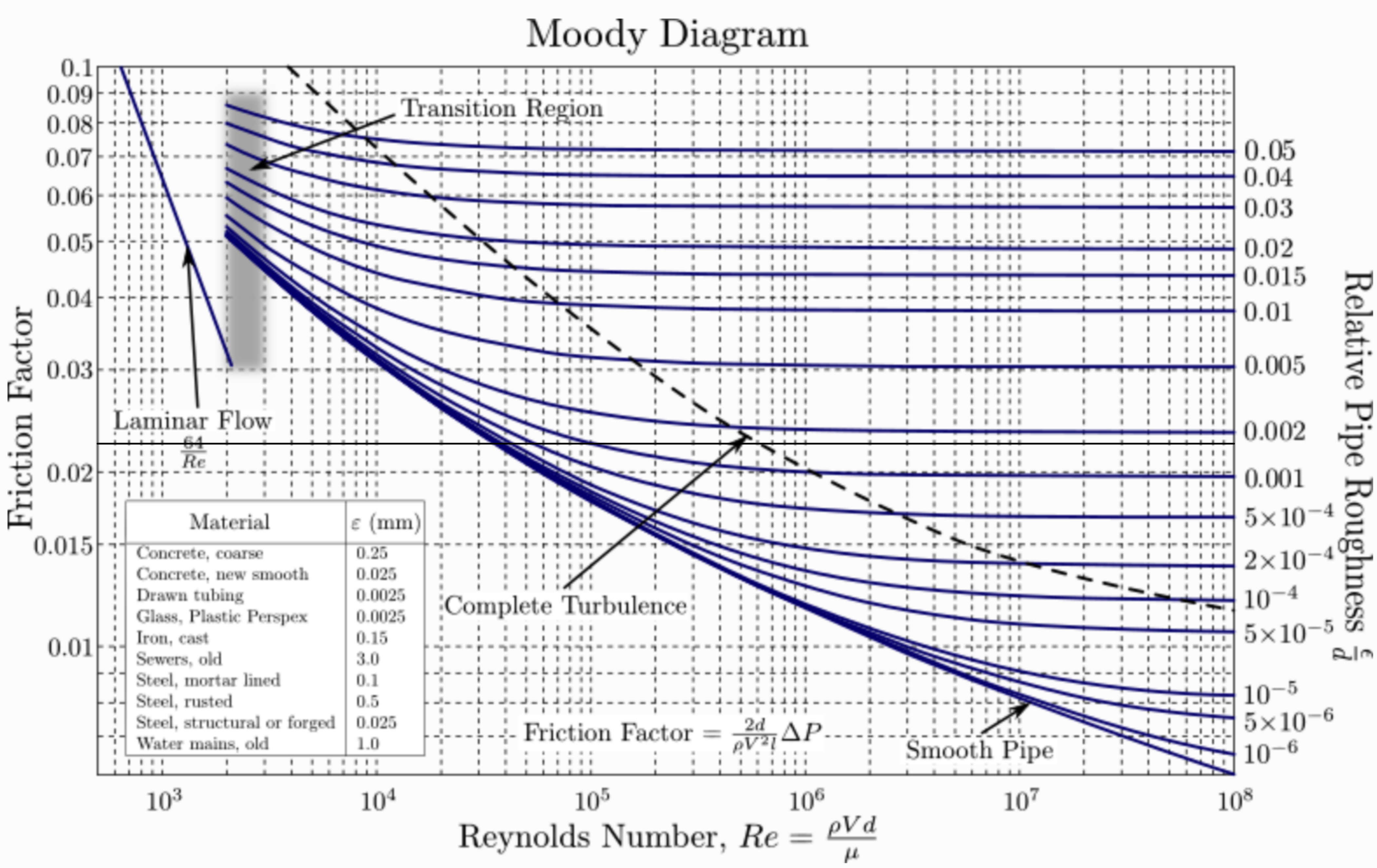
$L_1 = L_2 = L_4 = 10\,m \qquad L_3 = 15\,m$

$$Re = \frac{\rho V d}{\mu} = \frac{\rho V d}{\rho \nu} = \frac{Vd}{\nu}$$

$$\frac{V^2}{2g} = \frac{8Q^2}{g\pi^2 D^4}$$

Assuming completely turbulent flow:

$$\frac{\varepsilon}{D} = \frac{1.5}{80} = 0.01875$$



$f = 0.023$

3a) Major losses:

$$h_{L1} = h_{L2} = h_{L4} = f\frac{L}{D}\frac{V^2}{2g}$$

$$= 0.023\left(\frac{10}{0.08}\right)\frac{V^2}{2g}$$

$$= 2.875\left(\frac{V^2}{2g}\right)$$

$$h_{L3} = f\frac{L}{D}\frac{V^2}{2g} = 0.023\left(\frac{15}{0.08}\right)\frac{V^2}{2g}$$

$$= 4.3125\frac{V^2}{2g}$$

Minor losses:

$$h_{minor} = \sum K\frac{V^2}{2g}$$

$$= (0.5 + 2\times0.3 + 1.5 + 2\times0.3 + 3 + 0.4)\frac{V^2}{2g}$$

$$= 6.6\frac{V^2}{2g}$$

Total losses:

$$h_T = (4.3125 + 2\times2.875 + 6.6)\frac{V^2}{2g}$$

$$= 16.6625\frac{V^2}{2g}$$

3a) Assuming the velocity of the fluid stays constant throughout the system, and the pressure at the exit and entrance is atmospheric pressure:

$$\cancel{\frac{P_1}{\rho g}} + \cancel{\frac{V_1^2}{2g}} + z_1 - h_T = \cancel{\frac{P_2}{\rho g}} + \frac{V_2^2}{2g} + \cancel{z_2}$$

$$z_1 = \frac{V^2}{2g} + h_T$$

$$5 = \frac{V^2}{2g} + 16.6625\frac{V^2}{2g}$$

$$5 = 17.6625\left(\frac{8Q^2}{g\pi^2D^4}\right)$$

$$5 = 17.6625\left(\frac{8Q^2}{9.81\pi^2(0.08)^4}\right)$$

$$Q = 0.01184617848$$

$$\approx 0.0118\,m^3s^{-1}$$

b) Major losses:

Assuming completely turbulent flow and the fluid flow is split equally at junction B:

$$h_{L1} = h_{L2} = f\frac{L}{D}\frac{(\frac{1}{2}V)^2}{2g}$$

$$= \frac{1}{4}f\frac{L}{D}\frac{V^2}{2g}$$

$$= \frac{1}{4}(0.023)\left(\frac{10}{0.08}\right)\frac{V^2}{2g}$$

$$= 0.71875\frac{V^2}{2g}$$

3b) Major losses:

$$h_{major} = (4.3125 + 2\times 0.71875 + 2.875)\frac{V^2}{2g}$$
$$= 8.625\frac{V^2}{2g}$$

Minor losses:

$$h_{minor} = (0.5 + 2\times 0.3 + 1.5 + 4\times 0.3\times \tfrac{1}{4} + 2 + 3 + 0.4)\frac{V^2}{2g}$$
$$= 8.3\frac{V^2}{2g}$$

Total losses:

$$h_T = (8.625 + 8.3)\frac{V^2}{2g}$$
$$= 16.925\frac{V^2}{2g}$$

Assuming the pressure is atmospheric at the pipe exit:

$$\cancel{\frac{P_1}{\rho g}} + \cancel{\frac{V_1^2}{2g}} + z_1 - h_T = \cancel{\frac{P_2}{\rho g}} + \frac{V_2^2}{2g} + \cancel{z_2}$$

$$z_1 = h_T + \frac{V^2}{2g}$$

$$z_1 = 16.925\frac{V^2}{2g} + \frac{V^2}{2g}$$

$$z_1 = 17.925\left(\frac{8Q^2}{g\pi^2 D^4}\right)$$

$$5 = 17.925\left(\frac{8Q^2}{9.81\pi^2(0.08)^4}\right)$$

$$Q = 0.011759118 77$$
$$\approx 0.0118\,m^3s^{-1}$$

4a) $Q = 0.0628 \, m^3 s^{-1}$

$$V = \frac{Q}{A} = \frac{4Q}{\pi D^2} = \frac{4(0.0628)}{\pi (0.2)^2}$$

$$= 1.998986085 \, ms^{-1}$$

$$\approx 2.0 \, ms^{-1}$$

$$Re = \frac{VD}{\nu}$$

$$= \frac{2.0(0.2)}{1.0 \times 10^{-6}}$$

$$= 399797.217 > 4000$$

$$\approx 399797$$

$$\frac{\varepsilon}{D} = \frac{0.4 \times 10^{-3}}{20 \times 10^{-2}}$$

$$= 2 \times 10^{-3}$$

$$\frac{1}{\sqrt{f}} = -2\log\left(\frac{\frac{\varepsilon}{D}}{3.7} + \frac{2.51}{Re\sqrt{f}}\right)$$

$$\frac{1}{\sqrt{f}} = -2\log\left(\frac{2 \times 10^{-3}}{3.7} + \frac{2.51}{399797\sqrt{f}}\right)$$

Solving:

$$f = 0.0238783434$$

$$\approx 0.024$$

4a) Length of the pipe:

$$L = L_3 + h_1 + L_1 + L_2 + \frac{h_3}{\sin 30°}$$
$$= 3 + 3 + 10 + 2 + \frac{5}{\sin 30°}$$
$$= 28\,m$$

Major loss:

$$h_{major} = f \frac{L}{D} \frac{V^2}{2g}$$
$$= \frac{V^2}{2g}\left(0.024 \times \frac{28}{20 \times 10^{-2}}\right)$$
$$= 3.342968076 \frac{V^2}{2g}$$
$$\approx 3.34 \frac{V^2}{2g}$$

Minor loss:

$$h_{minor} = \Sigma K \frac{V^2}{2g}$$
$$= (3 + 0.2 + 0.1 + 1) \frac{V^2}{2g}$$
$$= 4.3 \frac{V^2}{2g}$$

Total losses:

$$h_T = (3.34 + 4.3) \frac{V^2}{2g}$$
$$= 7.642968076 \frac{V^2}{2g}$$
$$\approx 7.64 \frac{V^2}{2g}$$

4a) Energy equation from 1 to 2:

$$\cancel{\frac{p_1}{\rho g}} + \cancel{\frac{V_1^2}{2g}} + \cancel{z_1} + h_p - h_T = \cancel{\frac{p_2}{\rho g}} + \cancel{\frac{V_2^2}{2g}} + z_2$$

$$h_p = z_2 + h_T$$
$$= h_1 + h_2 + h_T$$
$$= 3 + 15 + 7.64\frac{(2.0)^2}{2\times 9.81}$$
$$= 25.84663502$$
$$\approx 25.85\ m$$

$$P = \rho g Q h_p = 1000\times 9.81\times 0.0628\times 25.85$$
$$= 15\,923.28474\ W$$
$$\approx 15.9\ kW$$

b) Energy equation from 1 to 2:

$$\frac{p_1}{\rho g} + \cancel{\frac{V_1^2}{2g}} + z_1 - h_L = \frac{p_s}{\rho g} + \frac{V_s^2}{2g} + z_s$$

$$\frac{p_{atm} - p_v}{\rho g} + z_1 - h_L = \frac{p_s - p_v}{\rho g} + \frac{V_s^2}{2g} + z_s$$

$$NPSH = \frac{p_{atm} - p_v}{\rho g} + z_1 - z_s - h_L$$

4b) $NPSH = \frac{P_{atm} - P_v}{\rho g} + z_1 - z_s - h_L$

$$= \frac{101.2\times10^3 - 2400}{1000\times9.81} + 0 - 3 - h_L$$

$$= 7.071355759 - h_L$$

$$\approx 7.07 - h_L$$

$$h_L = f\frac{L}{D}\frac{V^2}{2g} + \Sigma K\frac{V^2}{2g}$$

$$= \left(0.024\times\frac{3+3+10}{20\times10^{-2}} + 3 + 0.2\right)\frac{8Q^2}{\pi g D^4}$$

$$= 5.110267472\left(\frac{8(0.0628)^2}{9.81\pi(20\times10^{-2})^4}\right)$$

$$= 3.269746198\,m$$

$$\approx 3.27\,m$$

$$NPSH = 7.07 - 3.27$$

$$= 3.801609561\,m$$

$$\approx 3.8\,m$$

Assuming that the $NPSH_R = 0$, the piping system is not prone to cavitation, as $NPSH_A = 3.8\,m > 0\,m$.